

Disproof of TLMC Conjecture 00000000492

A “random hook length formula” expectation conjecture

TLMC submission

October 2, 2026

Abstract

Conjecture 00000000492 asserts that the expectation of the multiset of hook lengths of a uniformly random standard tableau, normalized as $(\sum \text{hooks})/n^2$, equals the closed form $(n^2 - 1)/3$. We show this is impossible: the left-hand side is capped at 1 for every tableau with n cells, while the conjectured value exceeds 1 for every $n \geq 3$. A concrete counterexample ($n = 3$, row shape) gives $2/3 \neq 8/3$. Alternative readings (square shapes, division by n) fail as well. Verdict: **FALSE**.

1 The conjecture

A random version of the hook length formula. *Conjecture:* the expectation of the multiset of hook lengths of a uniformly random standard tableau equals $(\sum \text{hooks})/n^2$, explicitly $(n^2 - 1)/3$.

We recall that for a Young diagram (partition) λ , the *hook length* of a cell c is the number of cells strictly to the right of c in its row, plus the number of cells strictly below c in its column, plus one. A basic but decisive observation is that the multiset of hook lengths depends only on the shape λ , not on the filling; hence the “expectation over uniformly random standard tableaux of the shape” is a constant attached to λ itself, and any universal bound on hook sums of a shape applies to the expectation verbatim.

2 The attack: a universal cap that the conjectured value exceeds

Fact 1 (Cap). *Let T be any standard tableau with n cells. Then every hook length of T is at most n , because the hook of a cell consists of the cell itself together with cells strictly right/below, and there are only n cells in total. Therefore*

$$\sum_{c \in T} h(c) \leq n \cdot n = n^2, \quad \text{so} \quad \frac{\sum h(c)}{n^2} \leq 1.$$

Fact 2 (The conjectured value overshoots the cap). *For every $n \geq 3$,*

$$\frac{n^2 - 1}{3} > 1,$$

since $n^2 - 1 > 3 \iff n^2 > 4$.

Combining Facts 1 and 2: for every tableau with $n \geq 3$ cells, of *any* shape,

$$\frac{\sum h(c)}{n^2} \leq 1 < \frac{n^2 - 1}{3},$$

so the conjecture is false for all $n \geq 3$, regardless of how the “random tableau” is distributed. The randomization is irrelevant: the conjectured value lies strictly above the largest value the left-hand side could possibly take.

3 Concrete counterexample

Counterexample 1 ($n = 3$, row shape (3)). *The shape (3) admits exactly one standard tableau, so the expectation is deterministic. Its hook lengths are $3, 2, 1$, with sum 6 . Then*

$$\frac{\sum h(c)}{n^2} = \frac{3 + 2 + 1}{9} = \frac{6}{9} = \frac{2}{3}, \quad \frac{n^2 - 1}{3} = \frac{9 - 1}{3} = \frac{8}{3},$$

and $2/3 \neq 8/3$. *The conjecture is false.*

4 Alternative readings also collapse

The statement is terse, so we check the plausible readings; all are false.

Square reading ($n \times n$ tableau, n^2 cells). The hook length of cell (i, j) of an $n \times n$ square is $(n - i) + (n - j) + 1$, whence

$$\sum h(c) = \sum_{i,j=1}^n ((n - i) + (n - j) + 1) = n^2(n - 1) + n^2 = n^3, \quad \frac{\sum h}{n^2} = n.$$

Setting $n = (n^2 - 1)/3$ gives $n^2 - 3n - 1 = 0$, whose discriminant 13 is not a perfect square; there is no integer solution, so the square reading fails for *every* $n \geq 1$. Example: the 3×3 square has hook multiset $\{5, 4, 4, 3, 3, 3, 2, 2, 1\}$, sum $27 = 3^3$, and $27/9 = 3 \neq 8/3$.

Divide-by- n reading ($(\sum h)/n = (n^2 - 1)/3$, n cells). Still $\sum h \leq n^2$, so $(\sum h)/n \leq n$, while $(n^2 - 1)/3 > n$ for every $n \geq 4$ (as $n^2 - 3n - 1 > 0$ there). Small cases fail directly: for $n = 3$, row shape gives $6/3 = 2 \neq 8/3$; for $n = 4$, row shape gives $10/4 = 5/2 \neq 5$.

Genuine randomization. The shape $(2, 1)$ ($n = 3$ cells) has two standard tableaux, both with the hook multiset $\{3, 1, 1\}$; the expectation is $E[\sum h] = 5$, and $E[\sum h]/9 = 5/9 \neq 8/3$, $E[\sum h]/3 = 5/3 \neq 8/3$. Randomization over tableaux of a fixed shape cannot move the hook-sum at all, and randomization over shapes cannot exceed the cap of Fact 1.

5 Boundary

- $n = 1$: LHS = 1 (either reading), conjectured value 0. Already false.
- $n = 2$: row shape gives $3/4$ (div. by n^2) or $3/2$ (div. by n) versus conjectured 1; the 2×2 square gives $8/4 = 2$ versus 1. False.
- $n \geq 3$: killed by the cap argument (div. by n^2), any shape; $n \geq 4$ killed by the cap under the divide-by- n reading; the square reading dies for every n .

6 Verification

The repository contains:

- `reproduce.py` — a standalone (stdlib-only) recomputation of every number above: hook multisets for shapes (3) , $(2, 1)$, $(3, 3, 3)$, the expectation for shape $(2, 1)$ via the hook length formula, the cap checks, and the discriminant argument.
- `lean4/` — a Lean 4 (v4.33.1, no Mathlib) development that concretizes the attack numbers as theorems proved by `rfl/decide` over `Nat`, including the counterexample $3 + 2 + 1 = 6$ with $6 \cdot 3 \neq 8 \cdot 9$ (i.e. $2/3 \neq 8/3$) and the overshoot $3 < n^2 - 1$ for $n \geq 3$. `Check.lean` audits every theorem with `#print axioms`; all are reported as *does not depend on any axioms* — zero axioms, zero `sorry`.